DMGNN: Diffusion-Enhanced Multimodal Graph Neural Network for HSI and LiDAR Classification

Changda Xing, Yifang Wu, Yongchang Xu, Cheng Wang, and Meiling Wang

***Abstract*—The fusion of hyperspectral images (HSIs) and light detection and ranging (LiDAR) data has become a cornerstone in remote sensing classification, leveraging complementary spectral and elevation information for accurate land cover classification. However, when modeling complex spatial relationships under limited labeled samples, the existing graph-based approaches often suffer from a "broken topology," i.e., a sparse and fragmented network structure that severely limits information flow. To address this limitation, in this work, we propose a novel diffusion-enhanced multimodal graph neural network (DMGNN), which takes the inherent topological advantages of graph representations to effectively model complex multimodal structures. A diffusion-enhanced multimodal graph construction module is first formulated to generate synthetic nodes that densify the fragmented topology, ensuring effective message passing even in sparse data regions. Second, a spectral clustering-guided pretraining strategy is introduced to mine the global cluster structure of the multimodal graph for superior parameter initialization. Third, a dynamic multimodal contrastive learning mechanism is constructed to enforce structural robustness and cross-modal alignment within the node embeddings. Fourth, an adaptive feature extraction and multiobjective optimization module is introduced to refine multiscale features and handle class imbalance. By flexibly utilizing topological structures, our method successfully addresses the scarcity of effective graph-based approaches. Extensive experiments on three widely used HSI and LiDAR datasets demonstrate that our proposed DMGNN outperforms the state-of-the-art methods. Notably, our method achieves an overall accuracy (OA) of 99.88% on Houston2013, 99.50% on Trento, and 95.14% on highly imbalanced Houston2018_sliced dataset. The codebase is now publicly available at https://github.com/11451419198106324/DMGNN**

***Index Terms*—Diffusion model, hyperspectral image (HSI) classification, light detection and ranging (LiDAR) images, multimodal datasets.**

I. Introduction

HYPERSPECTRAL images (HSI) provide rich, contiguous spectral signatures essential for fine-grained material identification across various remote sensing applications [1], [2]. Concurrently, light detection and ranging (LiDAR) has emerged as an indispensable active sensor that explicitly captures 3-D structural and elevation data [3]. Fusing these two modalities is exceptionally potent [4], [5]. While HSI excels at identifying spectral properties, LiDAR defines geometric boundaries [6]. This synergy is crucial for distinguishing spectrally similar but structurally different land-cover classes.

Traditional machine learning algorithms established a fundamental baseline for HSI classification but were often limited by their reliance on handcrafted feature extraction and lack of spatial context modeling [7], [8], [9], [10], [11]. While deep learning algorithms, particularly convolutional neural networks (CNNs), have significantly advanced HSI classification by extracting spatial–spectral representations [12], [13], [14], [15], [16], a fundamental limitation shared by these grid-based models is their reliance on regular Euclidean domains. Land cover distributions in remote sensing are inherently irregular, and complex spatial relationships often span beyond fixed convolutional kernels or rectangular windows.

To overcome the inherent limitations of grid-based processing, the graph neural networks (GNNs) have emerged as a powerful paradigm to model non-Euclidean spatial–spectral interactions [17], [18], [19], [20], [21], [22], [23]. By modeling pixels as nodes and their relationships as edges, GNNs can naturally capture long-range dependencies across irregular regions. Furthermore, unlike vision transformers (ViTs) [24], [25], [26], [27], [28], which typically require massive datasets to learn global context, GNNs leverage explicit topological constraints, making them highly effective under limited data conditions. Beyond 2-D applications, recent advancements in 3-D graph deep learning have demonstrated profound capabilities in processing laser point clouds and complex geometric topologies. For instance, the sophisticated graph-based frameworks combining 3-D spatial features and point clouds have achieved breakthrough performances in human–computer visual interaction and noncontact gesture recognition for intelligent rehabilitation [29], [30]. These

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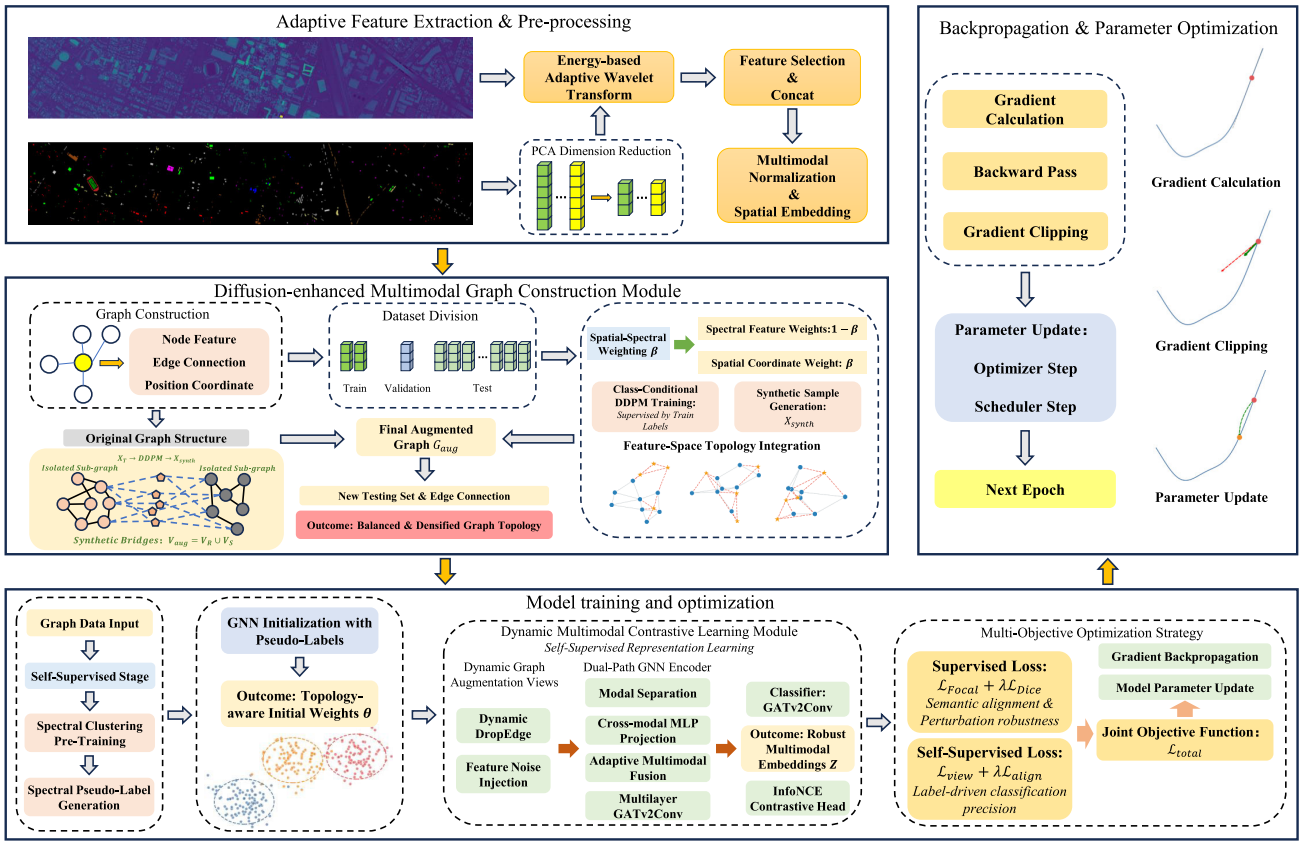


Fig. 1. Overview of the proposed DMGNN architecture.

successes strongly inspire the integration of graph structures with LiDAR data in remote sensing to resolve spatial ambiguities.

While multimodal fusion with LiDAR is critical, a major bottleneck remains the underutilization of graph representations [31], as the existing methods often rely on simple feature concatenation that ignores geometric topology [32], [33]. Even when standard graph methods are employed (GNNs and GATs), they typically depend on static graphs constructed purely from spectral similarity [34]. Under limited labeled samples, these static graphs suffer from problems, such as sparse, fragmented connections that sever message passing pathways.

To address the "broken topology" problem inherent in few-shot multimodal learning, we propose a diffusion-enhanced multimodal graph neural network (DMGNN). The proposed framework operates through a coordinated pipeline. First, adaptive wavelet feature extraction is applied to form compact, noise-suppressed node representations. Second, unlike standard GCNs that operate on fixed topologies, our framework utilizes a class-conditional diffusion model to generate synthetic nodes to act as structural bridges, which densifies the fragmented graph topology to facilitate information propagation. Third, a self-supervised spectral clustering-guided pretraining strategy is employed to initialize the GNN, providing a data-driven parameter foundation to mitigate overfitting under limited supervision. Finally, a dynamic multimodal contrastive learning mechanism synchronizes spatial–spectral semantics to reduce modality suppression. The network is jointly optimized with a hybrid focal-dice loss to handle long-tail distributions. The overall architecture is illustrated in Fig. 1.

The following are the main innovations in our work.

1) *Diffusion-Enhanced Multimodal Graph Construction Module:* To address the severe scarcity of annotated data in remote sensing classification, we integrate class-conditional diffusion models with spectral–spatial graph learning. Synthetic samples are dynamically generated and incorporated into the graph to balance the HSI-LiDAR class distribution.
2) *Spectral Clustering-Guided Pretraining:* To overcome the optimization instability associated with random GNN initialization under limited supervision in HSI and LiDAR multimodal classification, we employ spectral clustering, which provides structural pseudo-labels for a data-driven, topology-aware parameter initialization.
3) *Dynamic Multimodal Contrastive Learning:* To prevent dominant HSI spectral features from suppressing LiDAR geometric priors during fusion, we introduce a multimodal contrastive mechanism that strictly aligns cross-modal semantics.
4) *Adaptive Feature Extraction and Multiobjective Optimization:* To eliminate the spectral redundancy inherent in HSI and the high-frequency sensor noise in LiDAR data, we design an energy-driven adaptive wavelet transform. Coupled with a unified focal-dice objective,

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it comprehensively addresses class imbalance while enforcing spatial compactness in the final classification map.

The rest of this article is as follows. Section II presents the related works. Section III describes the methodology of our DMGNN. Section IV includes the experimental settings and results. The conclusions are given in Section V.

II. Related Works

A. Graph-Based and Generative Approaches

GNNs offer an alternative to grid-based models by mapping HSI data into non-Euclidean spaces, treating pixels as nodes and their similarities as edges. To address the massive computational cost of constructing full-batch adjacency matrices, Hong et al. [20] proposed miniGCN, which enables large-scale minibatch training and explicitly explores fusion strategies between CNN and GNN features. Concurrently, the generative adversarial networks (GANs) are frequently used to address data scarcity and domain shifts. For example, D3Net [35] utilizes a dual-confidence GAN to extract domain-invariant features. However, under few-shot scenarios, the existing GNN methods often suffer from topological degradation, where graphs built from sparse samples become highly fragmented. To address this, our DMGNN uses a class-conditional diffusion model to synthesize nodes that act as structural bridges, dynamically reconnecting the fragmented graph topology.

B. Multimodal Fusion of HSI and LiDAR

While HSI captures detailed spectral profiles, it often fails to distinguish materials with identical spectra but different elevations. Fusing HSI with LiDAR provides the necessary geometric context. Recent frameworks utilize self-attention and frequency-domain mechanisms for this fusion. For example, Fusion_HCT [5] combines a hierarchical CNN and transformer via cross-token attention, while MCFNet [36] applies discrete wavelet transforms (DWTs) for multiscale feature interaction. Graph-based fusion approaches, such as the graph-attention network GAMF [37], have also been proposed to model long-distance multimodal dependencies. The existing multimodal methods frequently suffer from modality suppression, where dominant HSI features overwhelm subtle LiDAR geometric cues and fail to actively optimize the topological structure itself. In contrast, our DMGNN combines adaptive wavelet extraction with dynamic contrastive learning to align cross-modal semantics and optimize the graph structure prior to message passing.

III. Methodology

A. Methodology Overview

In this section, we propose a novel multimodal graph-based classification framework (DMGNN), which is visually presented in Fig. 1. To fully exploit the topological potential of multimodal data, the framework integrates four coordinated components, including a diffusion-enhanced graph construction module to densify the topology and address data scarcity, a spectral clustering-guided pretraining strategy for robust parameter initialization, a dynamic multimodal contrastive learning mechanism to align features and enhance robustness, and an adaptive feature extraction and optimization module to handle class imbalance. The model's advantages will become apparent in subsequent experiments.

B. Diffusion-Enhanced Multimodal Graph Construction Module

In remote sensing classification, the severe scarcity and imbalance of annotated data remain critical bottlenecks [38], [39], [40]. While GNNs are highly effective at modeling non-Euclidean spatial–spectral relationships, their performance degrades catastrophically when operating on the sparse and fragmented graphs resulting from limited supervision. To address this data scarcity problem and fully leverage multimodal characteristics, we propose a diffusion-enhanced multimodal graph construction module. This module integrates a denoising diffusion probabilistic model (DDPM) [41] with spectral–spatial graph learning. By dynamically generating high-fidelity synthetic samples to balance the class distribution and seamlessly incorporating them into the graph, it reconstructs a dense, robust topology. Our approach follows three sequential stages: 1) constructing an initial spectral–spatial graph from real samples; 2) generating class-conditional synthetic samples via DDPM to balance the HSI-LiDAR class distribution; and 3) integrating them via class-specific K-NN connections to form the final augmented graph.

1) Spectral–Spatial Graph Construction: The initial graph $G_R = (V_R, E_R)$ is constructed from the set of N_R real, labeled pixels V_R. Each node $v_i \in V_R$ is defined by its spatial coordinates $\mathbf{p}_i = (r_i, col_i)$ and its fused multimodal feature vector $x_i \in \mathbb{R}^{d_{feat}}$ (where d_{feat} is the significantly reduced dimensionality after adaptive wavelet extraction). Specifically, x_i is derived by concatenating the processed HSI features h_i and LiDAR features l_i (obtained after PCA and the adaptive wavelet feature extraction)

$$x_i = h_i \oplus l_i. \tag{1}$$

To define the graph's edges, we must consider both feature similarity and spatial proximity. A key challenge is that these two data types exist on different scales. To address this, we first normalize the feature matrix X_R and the coordinate matrix P independently using Z-score normalization to give them zero mean and unit variance

$$\hat{X}_R = \frac{X_R - \mu_X}{\sigma_X}, \quad \hat{P} = \frac{P - \mu_P}{\sigma_P} \tag{2}$$

where μ_X and σ_X refer to the mean and standard deviation vectors of the feature matrix, respectively, and μ_P and σ_P refer to those of the coordinate matrix. Here, $\hat{X}_R$ and $\hat{P}$ represent the resulting standardized matrices. This prevents features with larger magnitudes from dominating the distance calculations. Furthermore, we introduce a tunable hyperparameter $\beta \in [0, 1]$ to explicitly control the balance between the two. As implemented, we create a joint feature representation f_i^{graph} for graph construction by performing a weighted concatenation

$$f_i^{graph} = (1 - \beta) \cdot \hat{x}_i \oplus \beta \cdot \hat{p}_i \tag{3}$$

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where $\oplus$ denotes the feature concatenation, $\hat{x}_i$ is the normalized multimodal feature vector, and $\hat{p}_i$ is the normalized coordinate vector for node v_i. A low β prioritizes feature similarity, connecting distant pixels with similar materials, while a high β prioritizes spatial contiguity. The initial adjacency matrix A_R (defining the edges E_R) is then built using a K-NN algorithm based on the Euclidean distance in this joint space f^{graph}.

2) Class-Conditional Diffusion Model: In HSI classification, the problem of small sample size remains a major challenge. While traditional augmentation methods, such as flipping, rotation, and noise injection, yield only minor variations and fail to model the true data manifold of rare classes, the DDPM model ensures training stability and generates high-fidelity synthetic representations [42], [43]. Our class-conditional diffusion framework leverages category labels to target minority classes (such as "stressed grass" or "artificial turf" in the Houston dataset), explicitly restoring balance to the HSI feature space.

Compared to advanced GANs [35], which rely on unstable min–max adversarial optimization and frequently suffer from mode collapse under extreme data scarcity, our DDPM approach offers distinct theoretical advantages for multimodal generation. In few-shot scenarios, a GAN discriminator easily overfits to the scarce real samples, causing the generator's gradients to vanish. Conversely, our DDPM optimizes a variational lower bound through a fixed forward Markov chain and a parameterized reverse denoising process. By explicitly matching the score function $\nabla_x \log p_t(x|c)$ of the data distribution rather than relying on an adversarial proxy, the diffusion model guarantees stable convergence. This ensures that the generated synthetic nodes capture the full diversity and continuous structural variance of the minority classes, preventing the network from merely memorizing limited training samples.

Unlike standard visual DDPMs that require massive datasets to learn high-dimensional image pixel distributions, our model operates directly on the 1-D fused feature vector $x_0 \in \mathbb{R}^{d_{feat}}$. This dimensionality reduction fundamentally transforms the generation task from complex image synthesis to low-dimensional manifold fitting. Consequently, instead of employing a computationally heavy U-Net, we utilize a lightweight ResNet-MLP [44] architecture. This strategic design drastically reduces the trainable parameter count and computational cost, ensuring stable convergence and preventing overfitting even under limited sample sizes.

Forward diffusion process [45] for noise injection: The forward process, denoted as q, is a fixed Markov chain that gradually destroys the structure of the data by adding Gaussian noise. Given a real data sample $x_0 \sim q(x)$ where $x_0 \in \mathbb{R}^{d_{feat}}$ represents the multimodal feature vector, we produce a sequence of latent variables $x_1, \ldots, x_T$ over T time steps. The transition probability at each step t is defined as

$$q(x_t|x_{t-1}) = \mathcal{N}\left(x_t;\ \sqrt{1-\beta_t}x_{t-1}, \beta_t I\right) \tag{4}$$

where $\mathcal{N}$ represents the Gaussian distribution and $\{\beta_t \in (0,1)\}_{t=1}^{T}$ is a predefined variance schedule. A key property of the diffusion process is that we can sample x_t at any arbitrary timestep t directly from x_0 using the reparameterization trick. Let $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$, the closed-form expression is

$$q(x_t|x_0) = \mathcal{N}\left(x_t;\ \sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t) I\right). \tag{5}$$

Thus, x_t can be expressed as

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, \quad \text{where } \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, I). \tag{6}$$

Reverse denoising process for conditional generation: The goal of the reverse process is to recover the original data x_0 from the pure noise $x_T \sim \mathcal{N}(\mathbf{0}, I)$. Since the true reverse distribution $q(x_{t-1}|x_t)$ is intractable, we approximate it using a parameterized neural network p_θ

$$p_\theta(x_{t-1}|x_t, c) = \mathcal{N}\left(x_{t-1}; \boldsymbol{\mu}_\theta(x_t, t, c), \boldsymbol{\Sigma}_\theta(x_t, t, c)\right). \tag{7}$$

Here, distinct from standard unconditional diffusion, our model is conditioned on the class label c. This allows the network to learn the specific spectral distribution of each land-cover category.

In our implementation, the network $\boldsymbol{\epsilon}_\theta$ (a ResNet-MLP architecture in our work) is trained to predict the noise component $\boldsymbol{\epsilon}$ that was added to x_0. The input into the network is a concatenation of the noisy feature x_t, the time embedding e_t, and the class embedding e_c

$$\hat{\boldsymbol{\epsilon}} = \boldsymbol{\epsilon}_\theta([x_t \oplus e_t \oplus e_c]). \tag{8}$$

To explicitly clarify data sufficiency, the exact training data for the diffusion model are the complete real labeled subset V_{train}. While 50 samples per class are generally insufficient to train a standard visual DDPM on high-dimensional images, our framework addresses this limitation. First, as established, the input is a highly compressed 1-D latent vector ($x_0 \in \mathbb{R}^{d_{feat}}$). By employing a lightweight ResNet-MLP architecture for denoising, we reduce the hypothesis space and parameter count compared to high-dimensional pixel-based DDPMs. Second, the noise prediction network $\boldsymbol{\epsilon}_\theta$ is trained jointly across the entire V_{train} pool (comprising $C \times N_c^{real}$ total samples) rather than on isolated per-class subsets. By sharing network weights across all C categories, $\boldsymbol{\epsilon}_\theta$ learns a universal, robust denoising mapping from the aggregated data, utilizing the learnable class embedding e_c merely as a semantic shift to guide the generation toward the specific conditional manifold $p(h, l \mid c)$.

The training objective is to minimize the simplified mean squared error (mse) loss between the actual noise and the predicted noise

$$\mathcal{L}_{\text{simple}}(\theta) = \mathbb{E}_{t,x_0,c,\boldsymbol{\epsilon}}\left[\left\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta\left(\sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t, c\right)\right\|^2\right]. \tag{9}$$

Generating high-fidelity samples from limited data requires strict structural regularization. For a specific land-cover class c, let h and l represent the HSI spectral features and LiDAR geometric features, respectively. Due to complex atmospheric and illumination variations, HSI features inherently exhibit high intraclass variance [Var($h|c$) is large]. Training a diffusion model directly on such a high variance space under extreme data scarcity typically leads to an unbounded score function $\nabla_x \log p_t(x|c)$, causing mode collapse and generation instability.

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However, by operating in a fused latent space $x = h \oplus l$, we explicitly introduce the modal characteristics of LiDAR. LiDAR provides deterministic physical and structural priors (e.g., elevation and scattering), resulting in a highly stable distribution with minimal variance [Var($l|c$) is small]. According to Bayes' theorem, the conditional joint probability distribution can be factorized as $p(h, l \mid c) = p(h \mid l, c)p(l \mid c)$. By conditioning the generation on the fused space, the structural stability of l anchors the diffusion trajectory. The theoretical condition for high-fidelity generation is satisfied because the conditional variance of the spectral features given the geometric priors, $\mathrm{Var}(h \mid l, c)$, is strictly bounded and significantly smaller than the marginal variance $\mathrm{Var}(h \mid c)$. This multimodal constraint ensures that the Lipschitz constant of the reverse diffusion score function remains bounded, thereby fulfilling the theoretical prerequisite for reliable, high-fidelity sample generation even when the empirical sample size is severely limited.

3) Augmented Graph Integration: Although the diffusion model [46] successfully generates high quality feature vectors (x_{synth}), GNNs require a topological structure to propagate information. A major challenge is that synthetic samples generated by diffusion lack inherent spatial coordinates, which means that they cannot be integrated into the graph using the spatial K-NN method used for real pixels. To resolve this, we propose a feature-space topology integration strategy. The core idea is to treat synthetic nodes as "bridges" or that connect real nodes of the same class, thereby densifying the intraclass connections and enhancing the discriminative power of the graph.

Synthetic Node Generation: Let N_c^{real} be the number of real training samples for class c, and let N^{target} be the desired number of samples per class. We generate $N_c^{synth} = \max(0, N^{\text{target}} - N_c^{\text{real}})$ synthetic feature vectors using the trained reverse process

$$x_0^{synth} \sim p_\theta(x_0|x_T, c) \tag{10}$$

which results in a set of synthetic nodes V_S with features X_S and the corresponding labels Y_S.

To prevent any feature scale or distribution mismatches, the diffusion model is explicitly trained on the latent space of the real nodes after the complete preprocessing pipeline. Consequently, the generated synthetic feature vectors $x_0^{synth} \in \mathbb{R}^{d_{feat}}$ inherently reside within the identical normalized distribution as the real data. Therefore, the synthetic feature matrix X_S strictly requires no secondary preprocessing and can be directly integrated with the real feature matrix X_R, mitigating feature shift and ensuring the numerical stability of subsequent GNN learning.

Based on statistical learning theory [47], diffusion generation mathematically compensates for data scarcity by reducing the empirical risk. Expanding a minority class c from N_c^{real} to N^{target} samples structurally compresses its generalization error bound from $\mathcal{O}(1/\sqrt{N_c^{real}})$ to $\mathcal{O}(1/\sqrt{N^{target}})$, which effectively prevents the decision boundary distortion typically caused by limited supervision.

Class-specific connectivity construction: To integrate a synthetic node $v_s \in V_S$ (with label y_s) into the graph, we search for its neighbors exclusively within the set of real nodes V_R that share the same label y_s. This class-specific constraint is crucial to prevent label information from being contaminated across class boundaries. The theoretical basis for this strict intraclass connectivity is rooted in the homophily assumption [48]. Permitting interclass edges (heterophily) for synthetic nodes would propagate conflicting semantic signals, directly exacerbating the oversmoothing problem in GNNs [49]. Strict intraclass connectivity ensures these nodes reinforce intraclass representations, preserving feature variance during gradient aggregation. We define the neighborhood $\mathcal{N}^*(v_s)$ using cosine similarity in the feature space to find the most semantically similar real samples. For a synthetic node v_s and a candidate real node v_r (where $y_s = y_r$), let x_s and x_r denote the corresponding feature vectors. The similarity is defined as

$$\mathrm{sim}(x_s, x_r) = \frac{x_s \cdot x_r}{\|x_s\|_2 \|x_r\|_2}. \tag{11}$$

The edge set for synthetic nodes, E_S, is constructed by connecting each v_s to its top-K nearest real neighbors

$$E_S = \left\{ (v_s, v_r) \,\middle|\, v_r \in \text{Top-K}_{v_j \in V_R, y_j = y_s} \left(\mathrm{sim}\left(x_s, x_j\right) \right) \right\}. \tag{12}$$

4) Final Graph Formulation: The final augmented graph $G_{\text{aug}} = (V_{\text{aug}}, E_{\text{aug}})$ unifies the original real data and the diffusion-generated data

$$V_{\text{aug}} = V_R \cup V_S \tag{13}$$

$$E_{\text{aug}} = E_R \cup E_S \cup E_S^T \tag{14}$$

where E_R represents the original spectral–spatial edges between real pixels, and E_S represents the semantic edges that connect the synthetic augmentation nodes to the real graph structure. This augmented topology allows the GNN to perform message passing on a balanced and enriched data distribution, significantly improving performance on tail classes.

Theoretically, the specific function of these synthetic nodes is to act as topological bridges that heal the "broken topology" of sparse classes. The impact of this densification can be rigorously quantified from the perspective of spectral graph theory. Injecting synthetic nodes strictly increases the algebraic connectivity of the graph, bounded by the Fiedler value λ_2 (the second smallest eigenvalue of the normalized graph Laplacian). A larger λ_2 mathematically indicates a denser topology that theoretically accelerates the random walk mixing time across the graph. Because this mixing time: the number of message passing steps required for information to propagate effectively is asymptotically bounded by $\mathcal{O}(1/\lambda_2)$, increasing λ_2 significantly broadens the message passing bandwidth. For the GNN model, this impact means that gradient flow can efficiently reach previously isolated real nodes without requiring deeper network architectures, thereby mitigating the oversmoothing problem while substantially improving the representation robustness of tail classes.

C. Spectral Clustering-Guided Pretraining

In HSI and LiDAR multimodal classification, training GNNs with limited labeled data often relies on random

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parameter initialization, which can lead to optimization instability and overfitting. To address this, we propose a spectral clustering-guided pretraining strategy. Instead of depending solely on scarce semantic labels, we use the fused multimodal features to oversegment the augmented graph. This process captures the underlying geometric structures of the data and generates structural pseudo-labels. By pretraining the GNN encoder with these pseudo-labels, we provide a data-driven, topology-aware initialization, which ensures more stable convergence during the supervised fine-tuning phase.

1) Spectral Clustering Formulation: We perform spectral clustering [50] on the nodes of the augmented graph G_{aug} to generate structural pseudo-labels. This process transforms the complex node connectivity into a low-dimensional embedding space, where the clusters are easily separable.

First, we utilize the affinity matrix $W \in \mathbb{R}^{|V_{aug}|\times|V_{aug}|}$, which is constructed using a nearest-neighbor similarity measure based on the Euclidean distance between nodes, and it represents the pairwise similarity between nodes in the multimodal feature space. Based on W, we compute the degree matrix D, which is a diagonal matrix, where each entry D_{ii} represents the accumulated weight of connections for node i

$$D_{ii} = \sum_{j=1}^{|V_{aug}|} W_{ij}. \tag{15}$$

The degree matrix captures the density of the graph at each node. Next, we compute the unnormalized Laplacian matrix L, defined as

$$L = D - W. \tag{16}$$

The Laplacian matrix is central to spectral graph theory, as it captures the local smoothness of the graph signal. However, since the node degrees in our graph can vary significantly (especially after adding synthetic nodes), we employ the normalized Laplacian matrix L_{sym} to ensure numerical stability and consistent clustering performance

$$L_{sym} = D^{-1/2}LD^{-1/2} = I_n - D^{-1/2}WD^{-1/2} \tag{17}$$

where I_n is the identity matrix. Normalization ensures that the clustering cuts are balanced, preventing the algorithm from simply separating isolated low-degree nodes from the rest of the graph.

The core step of spectral clustering is the Eigen-decomposition. We solve for the eigenvectors U and eigenvalues Λ of the normalized Laplacian

$$L_{sym}u_k = \lambda_k u_k. \tag{18}$$

We select the first K_{spec} eigenvectors corresponding to the smallest eigenvalues to form the spectral embedding matrix $U \in \mathbb{R}^{|V_{aug}|\times K_{spec}}$, where $|V_{aug}|$ is the node number of augmented graph. In this embedding space, the nonlinear manifold structure of the original high-dimensional data is unfolded, making the intrinsic clusters linearly separable. Finally, we apply the K-means algorithm on the rows of U to assign a discrete pseudo-label $\hat{y}_i$ to each node v_i

$$\hat{Y}_{spec} = \text{K-Means}(U), \quad \hat{y}_i \in \{1, \ldots, K_{spec}\}. \tag{19}$$

In our method, we set the hyperparameter $K_{spec} = 3 \times C$, where C is the number of real ground truth classes. This value is determined through an oversegmentation strategy to accommodate intraclass spectral variability. However, the dimensionality K_{spec} of the spectral embedding matrix U must be carefully bounded. If K_{spec} expands to approach the total node count $|V_{aug}|$, the spectral decomposition loses its dimensionality reduction capability and information bottleneck effect. In such a scenario, the resulting structural pseudo-labels $\hat{Y}_{spec}$ would fragment into meaningless isolated indices, providing no global topological prior, and our framework would indeed degenerate into a regular GCN with entirely random initialization. To systematically justify this heuristic and empirically demonstrate the optimal threshold, we conduct a rigorous sensitivity analysis of K_{spec} under extreme data scarcity, detailed in Section IV-E.

2) Pretraining Objective: With the generated pseudo-labels $\hat{Y}_{spec}$, we effectively turn the unsupervised data structure into a supervised learning signal. We attach a temporary projection head h_{pre} (a linear layer) to the GNN feature extractor f_θ. The objective is to train the GNN to map the raw input features X to these spectral clusters.

The pretraining loss function $\mathcal{L}_{pre}$ is formulated as the cross-entropy (CE) loss between the predicted cluster probabilities and the spectral pseudo-labels

$$\mathcal{L}_{pre} = -\frac{1}{B_{train}} \sum_{i=1}^{B_{train}} \sum_{k=1}^{K_{spec}} \mathbb{I}(\hat{y}_i = k) \log\left(p_{i,k}\right) \tag{20}$$

where B_{train} represents the size of the current training batch, $\mathbb{I}(\cdot)$ is the indicator function, and $p_{i,k}$ is the probability that node i belongs to cluster k, calculated via the softmax function

$$p_{i,k} = \frac{\exp\left(h_{pre}\left(f_\theta\left(x_i, E_{\text{aug}}\right)\right)k\right)}{\sum_{j=1}^{K_{spec}} \exp\left(h_{pre}\left(f_\theta\left(x_i, E_{\text{aug}}\right)\right)j\right)} \tag{21}$$

where $f_\theta(x_i, E_{\text{aug}})$ represents the node embedding generated by the GNN backbone. By minimizing $\mathcal{L}_{pre}$, the network weights θ are optimized to recognize the global cluster structure of the multimodal data. This process does not require ground truth annotations but ensures that the feature extractor is tuned to the underlying data manifold before the main training phase begins.

The proposed pretraining strategy provides an alternative to random initialization by explicitly leveraging the global graph topology. Although both real nodes V_R and synthetic nodes V_S have known semantic labels, these labels do not capture complex intraclass structural variations. Therefore, the structural pseudo-labels $\hat{Y}_{spec}$ are generated for all nodes in the augmented graph $V_{aug} = V_R \cup V_S$. By setting $K_{spec} = 3 \times C$, the spectral clustering functions as an oversegmentation process on the Laplacian of G_{aug}. Integrating HSI and LiDAR data allows geometric information to address spectral ambiguities and identify underlying geometric submanifolds. Pretraining the GNN encoder f_θ using $\hat{Y}_{spec}$ aligns the initial weights θ with this global topological distribution. Consequently, $\hat{Y}_{spec}$ acts as a structural regularizer, mitigating premature overfitting to limited semantic labels and establishing a structured initialization prior to supervised fine-tuning.

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D. Dynamic Multimodal Contrastive Learning

After graph construction and pretraining, the model enters the main representation learning phase. A critical issue in multimodal fusion is that simple feature concatenation often fails to fully exploit complementary information. Specifically, the network tends to overrely on the dominant HSI spectral features, thereby suppressing the critical geometric priors provided by LiDAR data. To prevent this modality suppression and mitigate GNN sensitivity to topological noise, we propose a dynamic multimodal contrastive learning module [51]. This module employs a dual-level mechanism. At the view level, it improves the model's robustness against structural perturbations. At the modality level, it aligns cross-modal representations to ensure that both spectral and geometric features are effectively balanced.

1) Dynamic Graph Augmentation Strategy: Unlike standard contrastive learning, we implement a dynamic augmentation strategy per epoch to prevent overfitting to specific noise patterns. Given the input graph $G_{\text{aug}} = (V_{\text{aug}}, E_{\text{aug}})$ with node features $X \in \mathbb{R}^{|V_{aug}| \times d_{feat}}$, we generate a corrupted view $\tilde{G}$ by randomly dropping edges [52] and injecting Gaussian noise δ

$$\tilde{G} = \left(X + \delta, \tilde{E}_{\text{aug}}\right) \tag{22}$$

where $\tilde{E}_{\text{aug}} \sim \text{DropEdge}(E_{\text{aug}}, p_{drop})$. The GNN encoder f_θ processes G_{aug} and $\tilde{G}$ to produce embeddings Z^{orig} and Z^{aug}.

Theoretically, dynamic graph augmentation enforces Lipschitz continuity to resist topological and feature noise [53]. Given feature perturbations bounded by $\|\delta\|_2 \leq \epsilon$ and an edge-missing probability p_{drop}, the contrastive alignment constrains the GNN's Lipschitz constant L_f. Consequently, the robustness boundary of the output embeddings ΔZ is strictly bounded by $\|\Delta Z\|_2 \leq L_f(\epsilon + \gamma p_{drop})$ (where γ scales topological shifts), mathematically preventing catastrophic manifold degradation.

2) Cross-Modal Semantic Alignment: This component specifically addresses the multimodal nature of the dataset. HSI data (h_i) captures material properties (e.g., distinguishing "grass" from "artificial turf"), while LiDAR data (l_i) capture geometric structure (e.g., distinguishing "ground" from "rooftops"). In the raw feature space, these two modalities are disjoint. To fuse them effectively, we construct a specific contrastive objective that forces the feature representations of the same pixel from different modalities to align in a shared latent space. We utilize specific projection heads $g_{hsi}(\cdot)$ and $g_{lidar}(\cdot)$ (implemented as MLPs) to map the modality-specific components of the raw input x_i into this shared space

$$z_{i,\text{hsi}} = g_{hsi}(h_i), \quad z_{i,\text{lidar}} = g_{lidar}(l_i). \tag{23}$$

By forcing $z_{i,\text{hsi}}$ to approximate $z_{i,\text{lidar}}$, LiDAR representations act as geometric anchors to correct ambiguous spectral features and vice versa.

3) InfoNCE Loss and Total Objective: We employ the InfoNCE [54] loss to optimize both objectives. For view-level contrast ($\mathcal{L}_{\text{view}}$), the positive pair comprises a node's embeddings from the original (z_i^{orig}) and augmented (z_i^{aug}) graphs, treating all other nodes in the batch B as negatives

$$\mathcal{L}_{\text{view}} = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\exp\left(\text{sim}\left(z_i^{\text{orig}}, z_i^{\text{aug}}\right)/\tau\right)}{\sum_{k=1,k\neq i}^{B} \exp\left(\text{sim}\left(z_i^{\text{orig}}, z_k^{\text{aug}}\right)/\tau\right)} \tag{24}$$

where $\text{sim}(\cdot,\cdot)$ is the cosine similarity and τ is the temperature parameter.

Similarly, the modal alignment loss ($\mathcal{L}_{\text{align}}$) forces the heterogeneous projections of the same node i to converge

$$\mathcal{L}_{\text{align}} = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\exp\left(\text{sim}\left(z_{i,\text{hsi}}, z_{i,\text{lidar}}\right)/\tau\right)}{\sum_{k=1,k\neq i}^{B} \exp\left(\text{sim}\left(z_{i,\text{hsi}}, z_{k,\text{lidar}}\right)/\tau\right)}. \tag{25}$$

The final contrastive objective $\mathcal{L}_{GCL}$ is the weighted sum of these topological and multimodal consistencies

$$\mathcal{L}_{GCL} = \mathcal{L}_{\text{view}} + \lambda_{\text{align}}\mathcal{L}_{\text{align}}. \tag{26}$$

Setting $\lambda_{\text{align}} = 0.5$, this joint optimization ensures the representations are robust to graph irregularities and semantically enriched by fusing complementary physical properties, effectively overcoming the limitations of static concatenation.

The modality-level alignment term is the key differentiator from single-modality HSI contrastive methods. By explicitly requiring HSI-derived representations to be close to their LiDAR-derived counterparts in the same physical object, our method injects geometric and elevation priors directly into the contrastive objective, producing embeddings that are both spectrally discriminative and geometrically consistent.

E. Adaptive Feature Extraction and Multiobjective Optimization

During multimodal remote sensing classification, the HSI data often suffer from severe spectral redundancy, while LiDAR elevation data usually contain high-frequency sensor noise. To eliminate these modality-specific artifacts and efficiently extract multiscale details, we design an energy-driven adaptive wavelet transform. Furthermore, remote sensing classification inherently faces extreme class imbalance. Coupled with the wavelet features, we introduce a unified focal-dice objective function that comprehensively addresses this long-tail distribution while enforcing spatial compactness in the final classification map.

1) Adaptive Wavelet Feature Extraction: While PCA reduces spectral redundancy, it often discards crucial high-frequency local textures. Conversely, traditional DWTs [55] use fixed decomposition levels, which may inadvertently retain sensor noise. To resolve this, we dynamically determine the optimal decomposition level for each feature channel. For a single channel $X_{img} \in \mathbb{R}^{H_{img} \times W_{img}}$, we apply the 2-D DWT using the *db*4 mother wavelet. At level j, the signal yields approximation $\mathbf{A}_j$ (low-frequency) and detail $\mathcal{D}_j = \{\mathbf{H}_j, \mathbf{V}_j, \mathbf{D}_j\}$ coefficients.

The total energy $E_{total}^{(j)}$ is calculated as

$$E_{total}^{(j)} = \|\mathbf{A}_j\|_2^2 + \sum_{\mathbf{d}\in\mathcal{D}_j} \|\mathbf{d}\|_2^2. \tag{27}$$

We define the high-frequency energy ratio R_j to quantify the structural information content in the detail subbands

$$R_j = \frac{\sum_{\mathbf{d}\in\mathcal{D}_j} \|\mathbf{d}\|_2^2}{E_{total}^{(j)}}. \tag{28}$$

The low frequencies represent the outline of an object, while the high frequencies represent texture or noise. If R_j is very

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small, it means that the high-frequency components of this layer are negligible or may even be just sensor noise. Iterating from $j = 1$ to J_{max}, if R_j falls below a threshold η (set to 0.05), further decomposition yields primarily noise. We stop at level j^* and construct the enhanced feature vector x_{wave} for each pixel i (extracting h_i and l_i at location $\mathbf{p}_i$)

$$x_{wave} = \text{Flatten}(\mathbf{A}_{j^*}) \oplus \text{Flatten}\left(\bar{\mathbf{D}}_{j^*}\right) \tag{29}$$

where $\bar{\mathbf{D}}_{j^*}$ is the mean of the detail coefficients. This ensures that the GNN receives a feature representation that is compact, denoised, and rich in multiscale spatial context.

From a signal processing perspective, this adaptive extraction functions as an energy-driven hard-thresholding filter [56] against high-frequency Gaussian sensor noise $\mathcal{N}(0, \sigma_n^2)$. By deliberately discarding detail subbands $\mathcal{D}_j$ when $R_j < \eta$, it truncates the spectrum exactly where the noise power overshadows structural signals, strictly bounding the residual noise variance propagated to downstream networks. This per-channel strategy is vital: HSI components require deeper decomposition for textures, while inherently smooth LiDAR channels saturate earlier.

2) Multiobjective Optimization Function: The standard CE loss assigns equal weight to all classes and pixels, which limits its effectiveness on datasets with severe class imbalance and spatial consistency requirements (such as Houston 2018). To address this, we formulate a hybrid loss function combining focal loss [57] and dice loss [58]. During training, the synthetic nodes V_S and their generated labels Y_S are incorporated as ground truth targets in the supervised loss computation. By participating directly in the gradient updates of the classifier, the sample index i in the supervised losses encompasses both the real labeled nodes and the synthetic nodes, effectively rebalancing the gradient contributions across classes. To prevent easy-to-classify samples from overwhelming the gradient, focal loss dynamically focuses on hard, misclassified examples. Let $p_{i,c}$ be the predicted probability that pixel i belongs to class c

$$\mathcal{L}_{Focal} = -\frac{1}{B}\sum_{i=1}^{B}\sum_{c=1}^{C} y_{i,c}\alpha_c(1 - p_{i,c})^{\gamma}\log(p_{i,c}) \tag{30}$$

where $y_{i,c} \in \{0, 1\}$, $\gamma = 4.0$ is the focusing parameter, and α_c balances class frequencies.

To complement pixelwise classification with spatial structure optimization, we employ the generalized dice loss to maximize the intersection over union (IoU) of the predicted regions

$$\mathcal{L}_{Dice} = 1 - \frac{1}{C}\sum_{c=1}^{C}\frac{2\sum_{i=1}^{B} p_{i,c}y_{i,c} + \epsilon}{\sum_{i=1}^{B} p_{i,c} + \sum_{i=1}^{B} y_{i,c} + \epsilon} \tag{31}$$

where ϵ improves numerical stability, and averaging across C prevents majority class dominance.

Total objective function: the final objective function $\mathcal{L}_{total}$ integrates the supervised classification objectives with the self-supervised contrastive learning objective to train the entire network end-to-end

$$\mathcal{L}_{total} = \underbrace{(\mathcal{L}_{Focal} + \lambda_{Dice}\mathcal{L}_{Dice})}_{\text{Supervised Classification}} + \underbrace{\lambda_{GCL}\left(\mathcal{L}_{view} + \lambda_{align}\mathcal{L}_{align}\right)}_{\text{Self-supervised Contrastive}} \tag{32}$$

where setting hyperparameters $\lambda_{Dice} = 0.3$, $\lambda_{GCL} = 0.1$, and $\lambda_{align} = 0.5$, and this joint optimization produces discriminative features, coherent maps, and robust multimodal representations, preventing dominant HSI features from suppressing LiDAR geometric cues.

In summary, we propose DMGNN, which tightly couples four coordinated components around the global optimization of the GNN encoder f_θ, rather than functioning as isolated steps. First, the adaptive feature extraction module applies energy-driven wavelet decomposition to generate noise-suppressed features x_{wave}. Based on these representations, the diffusion module generates class-conditional synthetic nodes to construct a densified graph topology E_{aug}. Subsequently, spectral clustering-guided pretraining utilizes the Laplacian of E_{aug} to initialize the GNN weights θ. During the end-to-end training phase, the multimodal node embeddings $Z = f_\theta(x_{wave}, E_{aug})$ are optimized via the unified objective function $\mathcal{L}_{total}$. Specifically, dynamic contrastive learning ($\mathcal{L}_{view}$ and $\mathcal{L}_{align}$) provides self-supervised regularization for representation consistency, while the multiobjective loss ($\mathcal{L}_{Focal}$ and $\mathcal{L}_{Dice}$) optimizes classification accuracy and spatial coherence.

IV. Experiment and Results

A. Description of Datasets

1) Houston2013: The Houston2013 dataset, acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS), covers a complex urban area in Houston, Texas. It consists of 144 spectral bands spanning the 400–2500-nm range with a high spatial resolution of approximately 2.5 m. The dataset captures diverse land-cover types, including buildings, roads, vegetation, and water, providing rich spectral and spatial details for urban classification.

2) Trento: The Trento dataset is a classic benchmark for multimodal remote sensing classification. It consists of HSI acquired by the AISA Eagle sensor and LiDAR digital surface models (DSMs) acquired by the Optech ALTM 3100EA sensor. Both modalities are rigorously registered, possessing a high spatial resolution of 1 m and an image size of 600×166 pixels. The HSI data contain 63 spectral channels in the wavelength range of 402.89–989.09 nm, with a spectral resolution of 9.2 nm.

3) Houston2018_Sliced: To address the high computational memory costs of the original data while preserving classification complexity, we constructed the Houston2018_sliced dataset. Specifically, we defined the complexity of a region based on the diversity of its land-cover distribution, quantified by Shannon entropy [59]

$$E(x) = -\sum_{c\in\mathcal{C}} p_c(x)\log_2 p_c(x) \tag{33}$$

where $p_c(x)$ represents the empirical probability of class c within a sliding window $\mathcal{R}_x$ of width $W'_{img} = \lfloor W_{img}/5 \rfloor$. By maximizing this score ($x^* = \text{argmax}E(x)$), we selected the region starting at $x^* = 903$ (entropy: 3.3521) as the most texturally complex and information-rich area.

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TABLE I
COMPARATIVE EXPERIMENTAL CLASSIFICATION RESULTS ON THE HOUSTON2013 DATASET

Metric	Bi-LSTM	DSFormer	Fusion_HCT	GAMF	MCFNet	MTFFN	miniGCN	DMGNN (Ours)
OA(%)	54.68 ± 3.95	98.57 ± 0.46	98.73 ± 0.23	95.31 ± 0.21	96.29 ± 1.70	84.03 ± 1.69	94.82 ± 0.35	**99.88 ± 0.02**
AA(%)	52.79 ± 3.58	98.78 ± 0.46	98.94 ± 0.18	96.08 ± 0.17	91.91 ± 3.81	85.94 ± 1.47	95.69 ± 0.28	**99.91 ± 0.02**
$\kappa\times100$	50.87 ± 4.29	98.45 ± 0.51	98.62 ± 0.25	94.92 ± 0.22	95.98 ± 1.87	82.75 ± 1.82	94.38 ± 0.38	**99.87 ± 0.02**
Class 1	61.22 ± 3.08	98.49 ± 0.80	99.43 ± 0.42	94.45 ± 3.24	96.71 ± 1.41	89.90 ± 6.37	98.12 ± 0.99	**99.82 ± 0.00**
Class 2	59.05 ± 10.58	98.84 ± 0.33	99.63 ± 0.26	95.46 ± 3.37	99.79 ± 0.26	93.28 ± 4.25	99.03 ± 0.56	**99.82 ± 0.00**
Class 3	72.86 ± 11.63	**100.00 ± 0.00**	99.84 ± 0.00	98.81 ± 1.99	99.89 ± 0.15	91.79 ± 8.95	**100.00 ± 0.00**	**100.00 ± 0.00**
Class 4	87.96 ± 5.61	99.97 ± 0.05	99.66 ± 0.31	98.23 ± 0.37	99.58 ± 0.47	95.46 ± 2.92	98.54 ± 0.37	**100.00 ± 0.00**
Class 5	76.24 ± 2.25	**100.00 ± 0.00**	**100.00 ± 0.00**	99.48 ± 0.44	**100.00 ± 0.00**	96.52 ± 1.50	99.82 ± 0.07	**100.00 ± 0.00**
Class 6	56.92 ± 5.29	99.56 ± 0.45	**100.00 ± 0.00**	95.69 ± 3.53	45.32 ± 31.48	88.96 ± 1.21	99.43 ± 0.81	**100.00 ± 0.00**
Class 7	42.31 ± 3.92	97.80 ± 3.21	98.51 ± 1.05	96.37 ± 0.69	99.38 ± 0.57	78.66 ± 13.10	86.97 ± 0.87	99.85 ± 0.09
Class 8	51.63 ± 4.96	93.97 ± 1.87	93.73 ± 2.94	94.60 ± 0.43	98.72 ± 0.82	76.04 ± 3.86	91.99 ± 1.80	**100.00 ± 0.00**
Class 9	30.03 ± 1.16	97.95 ± 0.62	96.76 ± 0.75	93.05 ± 1.60	97.48 ± 0.91	70.62 ± 11.58	84.69 ± 3.18	**99.91 ± 0.00**
Class 10	48.50 ± 5.93	99.65 ± 0.36	98.87 ± 0.89	90.60 ± 1.58	99.46 ± 0.45	82.60 ± 10.92	92.45 ± 4.32	**99.75 ± 0.04**
Class 11	41.29 ± 3.91	99.24 ± 0.58	99.77 ± 0.18	94.35 ± 2.39	**100.00 ± 0.00**	55.13 ± 20.17	94.07 ± 1.67	99.91 ± 0.00
Class 12	5.63 ± 1.81	98.00 ± 0.47	99.02 ± 0.43	91.17 ± 2.55	99.10 ± 0.92	80.10 ± 9.16	95.69 ± 1.55	**99.57 ± 0.23**
Class 13	41.64 ± 2.73	98.74 ± 2.19	98.89 ± 0.85	96.67 ± 1.23	64.61 ± 45.13	90.26 ± 5.05	94.57 ± 2.43	**100.00 ± 0.00**
Class 14	40.91 ± 8.63	**100.00 ± 0.00**	**100.00 ± 0.00**	99.80 ± 0.11	78.61 ± 31.36	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
Class 15	0.00 ± 0.00	99.52 ± 0.51	**100.00 ± 0.00**	98.72 ± 1.32	**100.00 ± 0.00**	99.79 ± 0.30	99.93 ± 0.09	**100.00 ± 0.00**

The resulting dataset features spatial dimensions of 1202 × 953 pixels and comprises 50-band HSI data co-registered with single-band LiDAR data at a 1-m resolution. It encompasses 15 remapped land-cover classes ranging from "stressed grass" to "stadium seats." A defining characteristic of this subset is its severe class imbalance (e.g., 144 pixels for "water" versus 73 000+ for "nonresidential buildings"), making it a rigorous benchmark for evaluating the robustness of multimodal classification models in dense urban environments.

B. Experimental Settings

1) Implementation Details: The proposed framework is implemented in Python 3.10 using PyTorch, PyG, and Scikit-learn, running on a single NVIDIA GPU with AMP. The main model is trained for 200 epochs using AdamW (lr = 1×10^{-3} and weight decay = 3×10^{-4}) and a cosine annealing scheduler, while the diffusion model is pretrained for 300 epochs (lr = 1×10^{-4}). We partition the datasets using a fixed-sample strategy per class for the training subsets (V_{train}: 50 for Houston2013, 80 for Trento, and 100 for Houston2018_sliced), 50 for validation (V_{val}), and the remainder for testing (V_{test}). To strictly prevent information leakage, synthetic nodes V_S are generated conditioned exclusively on V_{train} labels. During validation and testing, V_S remains in the overall graph solely to support structural message passing, thereby preventing topological distribution shifts, but the GNN metrics are evaluated strictly on the real unseen nodes in V_{val} and V_{test}. Maintaining these nodes ensures that the graph density remains consistent across all phases, effectively neutralizing topological distribution shifts. By rigorously reconstructing the true conditional manifold $p(h, l|c)$, V_S effectively bridges the sparsity gap without introducing feature-level distribution shifts between the training and inference phases.

We evaluate the performance of DMGNN using four main indicators: overall accuracy (OA), average accuracy (AA), kappa coefficient (κ), and the confusion matrix, and we included the mean and standard deviation for each category.

2) Comparison Methods: To evaluate the effectiveness of our proposed framework in exploiting topological potential compared to diverse architectures, we compare it with seven state-of-the-art classification methods. Bi-LSTM [60]: utilizes a bidirectional recurrent network with spatial–spectral attention to capture sequential spectral dependencies; DSFormer [61]: employs a dual selective fusion mechanism to adaptively adjust receptive fields and filter informative tokens; Fusion_HCT [5]: integrates a hierarchical CNN and a transformer through cross-token attention to fuse HSI and LiDAR features; GAMF [37]: constructs a heterogeneous graph with Gaussian tokenization to model long-range dependencies between multimodal features; MCFNet [36]: incorporates DWT to capture frequency-domain details alongside spatial features; MTFFN [62]: adopts a dual-branch transfer learning strategy to leverage spatial and spectral knowledge from heterogeneous source domains; MiniGCN [20]: allows training large scale GCNs in a minibatch fashion to extract spatial–spectral features without retraining networks for out-of-sample data. These benchmarks cover diverse paradigms, including RNNs, transformers, hybrid architectures, and multimodal graph methods, which provide a comprehensive evaluation of the proposed method's capability in multimodal feature learning.

C. Performance Comparison and Analysis

1) Overall Metrics Analysis: According to Tables I–III, the experimental results across the Houston2013, Trento, and Houston2018_sliced datasets demonstrate that DMGNN consistently outperforms the state-of-the-art methods, achieving the highest OA, AA, and κ. While DMGNN achieves near perfect accuracy on Houston2013 (99.88% OA) and Trento (99.50% OA), its superiority is most pronounced on the challenging Houston2018_sliced dataset. Here, DMGNN achieves an OA of 95.14%, establishing a significant lead over competitive baselines, such as MCFNet (87.62%), DSFormer (86.12%), GAMF (86.86%), and miniGCN (84.27%), which struggle with the dataset's complexity and class imbalance.

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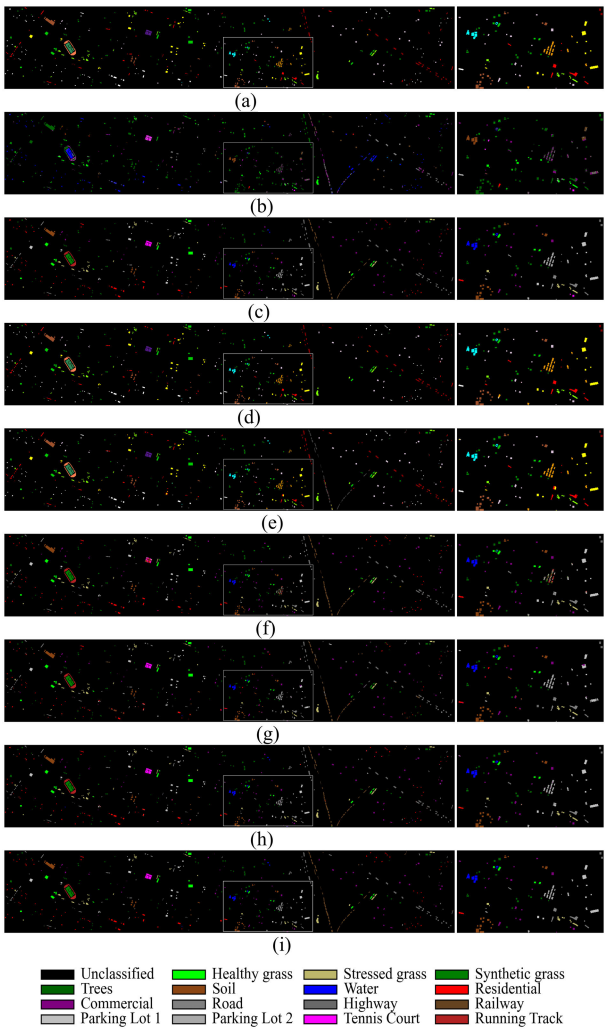


Fig. 2. Comparative experimental results on the Houston2013 dataset. (a) Ground truth. (b) Bi-LSTM. (c) DSFormer. (d) Fusion_HCT. (e) GAMF. (f) MCFNet. (g) MTFFN. (h) miniGCN. (i) DMGNN (ours).

In addition, we provide the confusion matrix evaluation in Section B of *Supplementary Materials*, with the results shown in Figs. S1–S3. As can be found, our DMGNN demonstrates exceptional precision and recall on extreme minority classes, producing almost no false positives or false negatives.

These metrics directly validate the advantage of our graph-based architecture in processing multimodal data with limited labeled samples. The significant performance gap in minority classes, such as Class 11 in Houston2018_sliced dataset, where DMGNN achieves 99.82% compared to Bi-LSTM's 0.01% and MTFFN's 7.79%, confirms the effectiveness of the proposed graph topology. By treating pixels as nodes, DMGNN facilitates efficient message passing between scarce labeled samples and unlabeled neighbors, allowing geometrical and spectral features to propagate effectively even in data-sparse regions where traditional grid-based or sequential models fail.

2) Visualization Results Analysis: Figs. 2–4 provide the classification maps for comparative experiments. In Fig. 4, the Houston2018_sliced dataset highlights significant performance disparities across the methods, using the ground truth [Fig. 4(a)] as the standard. Bi-LSTM [Fig. 4(b)] exhibits severe "salt-and-pepper" noise, particularly in large homogeneous regions, such as stadium seats and parking lots, resulting in a fragmented appearance. MTFFN [Fig. 4(g)] performs the worst, failing to classify large sections of the image (visible as missing black voids) and showing poor continuity in road networks. Transformer-based methods, such as DSFormer [Fig. 4(c)] and Fusion_HCT [Fig. 4(d)], improve regional coherence but still suffer from edge blurring around complex building boundaries. Graph-based methods, such as GAMF [Fig. 4(e)] and miniGCN [Fig. 4(h)], show decent structural preservation but misclassify minor classes embedded within the urban clutter.

In contrast, our DMGNN [Fig. 4(i)] produces the highest precision map by successfully suppressing noise in smooth areas while preserving the sharp geometric edges of buildings and narrow sidewalks. Furthermore, the visual results on the Houston2013 dataset (Fig. 2) explicitly clarify a critical difference in spatial feature handling. While grid-based architectures, such as Fusion_HCT [Fig. 2(d)], may appear macroscopically smooth due to inherent low-pass filtering, they blur sharp geometric edges. In contrast, the dense visual details in the DMGNN classification map [Fig. 2(i)] are not misclassified fragmented noise, as seen in MTFFN, but rather highly precise delineations of high-frequency spatial structures. Similar superior trends and noise reduction capabilities are observed in the Trento dataset (Fig. 3), underscoring the consistent robustness of DMGNN.

These visual results are in agreement with the proposed methodological framework. The exceptional spatial smoothness and lack of fragmentation in the DMGNN classification maps, particularly in the Houston2018_sliced dataset, are directly attributed to the diffusion-enhanced multimodal graph construction. By generating synthetic nodes to bridge disconnected regions, our graph method ensures effective message passing even in sparse areas. Furthermore, the sharp delineation of building boundaries is a result of dynamic multimodal contrastive learning, which aligns HSI spectral features with LiDAR geometric constraints, effectively resolving the misclassification problem. Finally, the integration of dice loss within the multiobjective optimization module explicitly penalizes discontinuous predictions, enforcing the spatial compactness observed in our final results.

D. Ablation Study

To evaluate the contributions of individual components to the performance of DMGNN, we conducted a systematic ablation study on the Houston2013 dataset. We established a basic multimodal GNN as the primary baseline and evaluated the model's robustness by selectively removing key modules. The experimental settings include: 1) baseline (HSI), a basic GNN using only hyperspectral data; 2) baseline (HSI + LiDAR), a standard multimodal GNN without advanced enhancements; 3) w/o LiDAR, the proposed DMGNN framework operating on HSI data; 4) w/o diffusion, removing the diffusion-enhanced graph construction to test the impact of synthetic sample generation; 5) w/o pretrain, excluding the spectral clustering-guided initialization; 6) w/o GCL, removing the

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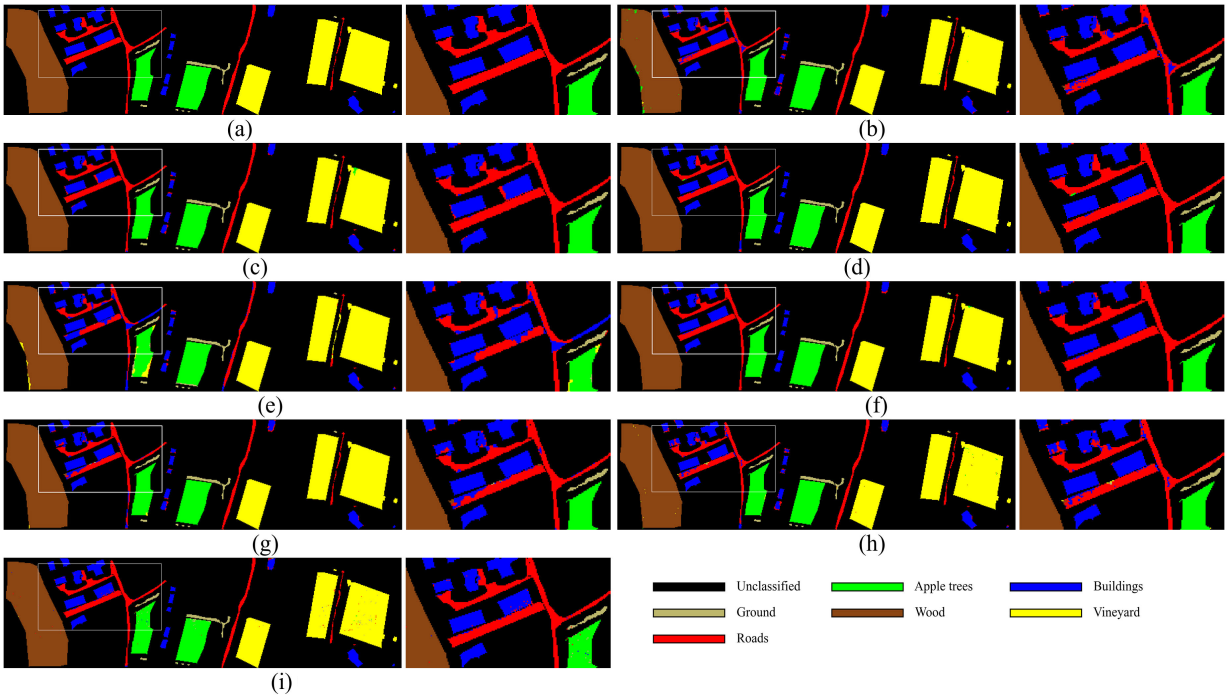


Fig. 3. Comparative experimental results on the Trento dataset. (a) Ground truth. (b) Bi-LSTM. (c) DSFormer. (d) Fusion_HCT. (e) GAMF. (f) MCFNet. (g) MTFFN. (h) miniGCN. (i) DMGNN (ours).

TABLE II
COMPARATIVE EXPERIMENTAL CLASSIFICATION RESULTS ON THE TRENTO DATASET

Metric	Bi-LSTM	DSFormer	Fusion_HCT	GAMF	MCFNet	MTFFN	miniGCN	DMGNN (Ours)
OA(%)	92.57 ± 2.55	98.92 ± 0.10	99.46 ± 0.06	98.18 ± 0.79	99.45 ± 0.04	97.56 ± 1.04	98.15 ± 0.07	**99.50 ± 0.07**
AA(%)	89.54 ± 2.52	98.44 ± 0.11	99.03 ± 0.20	97.38 ± 1.16	98.98 ± 0.16	96.01 ± 1.77	96.88 ± 0.19	**99.28 ± 0.12**
$\kappa\times100$	90.10 ± 3.43	98.55 ± 0.13	99.28 ± 0.08	97.57 ± 1.24	99.26 ± 0.05	96.75 ± 1.39	97.52 ± 0.10	**99.34 ± 0.08**
Class 1	88.62 ± 7.48	99.70 ± 0.10	99.04 ± 0.17	97.89 ± 1.84	99.32 ± 0.39	99.76 ± 0.11	**99.88 ± 0.15**	99.01 ± 0.14
Class 2	86.51 ± 3.23	96.64 ± 1.36	98.54 ± 0.90	95.51 ± 4.04	**98.65 ± 0.91**	94.27 ± 4.82	91.97 ± 1.87	98.56 ± 0.64
Class 3	84.00 ± 4.34	**100.00 ± 0.00**	99.08 ± 1.13	97.61 ± 1.91	98.91 ± 0.85	99.27 ± 1.04	98.95 ± 0.82	**100.00 ± 0.00**
Class 4	98.31 ± 1.08	99.99 ± 0.01	**100.00 ± 0.00**	99.97 ± 0.01	**100.00 ± 0.00**	99.96 ± 0.03	99.93 ± 0.05	99.87 ± 0.10
Class 5	92.88 ± 5.34	99.48 ± 0.18	**99.98 ± 0.03**	98.80 ± 0.89	99.96 ± 0.04	99.89 ± 0.10	99.72 ± 0.04	99.97 ± 0.03
Class 6	86.94 ± 4.50	94.85 ± 1.82	97.55 ± 0.95	93.64 ± 6.01	97.06 ± 1.29	82.91 ± 13.62	90.86 ± 1.79	**98.25 ± 0.64**

TABLE III
COMPARATIVE EXPERIMENTAL CLASSIFICATION RESULTS ON THE HOUSTON2018_SLICED DATASET

Metric	Bi-LSTM	DSFormer	Fusion_HCT	GAMF	MCFNet	MTFFN	miniGCN	DMGNN(Ours)
OA(%)	88.10 ± 1.03	86.12 ± 3.69	87.69 ± 0.93	86.86 ± 0.72	87.62 ± 0.52	60.55 ± 19.70	84.27 ± 0.88	**95.14 ± 0.12**
AA(%)	82.39 ± 1.92	90.97 ± 1.73	91.26 ± 0.22	89.50 ± 0.34	88.48 ± 3.10	58.07 ± 18.12	88.04 ± 0.24	**96.99 ± 0.07**
$\kappa\times100$	86.49 ± 1.18	84.45 ± 4.10	86.14 ± 1.02	85.22 ± 0.79	86.04 ± 0.52	56.91 ± 20.68	82.35 ± 0.95	**94.51 ± 0.13**
Class 1	85.04 ± 1.21	91.71 ± 6.41	91.47 ± 3.16	90.09 ± 3.55	88.45 ± 2.77	54.67 ± 23.44	86.14 ± 2.42	**98.03 ± 0.24**
Class 2	91.44 ± 1.68	91.62 ± 2.65	86.09 ± 2.41	**100.00 ± 0.00**	91.50 ± 0.83	78.62 ± 20.84	88.75 ± 0.72	91.67 ± 0.52
Class 3	99.67 ± 0.09	99.97 ± 0.02	**100.00 ± 0.00**	98.30 ± 1.84	**100.00 ± 0.00**	66.00 ± 46.47	99.83 ± 0.08	**100.00 ± 0.00**
Class 4	95.82 ± 1.42	97.49 ± 2.63	98.76 ± 0.57	92.96 ± 2.36	97.13 ± 1.05	69.97 ± 28.48	97.71 ± 0.95	**99.75 ± 0.08**
Class 5	94.04 ± 0.36	98.30 ± 0.82	98.84 ± 0.56	**100.00 ± 0.00**	94.31 ± 2.97	62.16 ± 43.79	97.21 ± 1.13	99.52 ± 0.11
Class 6	77.30 ± 15.90	**100.00 ± 0.00**	**100.00 ± 0.00**	97.70 ± 0.30	**100.00 ± 0.00**	32.61 ± 44.10	0.00 ± 0.00	**100.00 ± 0.00**
Class 7	98.44 ± 0.24	96.32 ± 3.10	98.75 ± 0.37	95.74 ± 1.05	99.53 ± 0.29	90.55 ± 2.81	97.29 ± 0.15	**99.92 ± 0.02**
Class 8	96.61 ± 0.59	89.93 ± 0.51	94.54 ± 1.21	71.39 ± 1.74	95.68 ± 1.82	63.41 ± 33.25	89.17 ± 0.76	**98.17 ± 0.16**
Class 9	64.56 ± 4.15	81.84 ± 2.41	71.57 ± 6.16	71.15 ± 5.71	59.35 ± 28.08	48.08 ± 19.40	67.85 ± 5.09	**94.12 ± 0.64**
Class 10	66.25 ± 4.15	75.46 ± 1.48	66.42 ± 3.01	78.80 ± 9.75	73.44 ± 2.71	52.34 ± 26.95	68.84 ± 2.52	**88.79 ± 0.80**
Class 11	0.01 ± 0.01	87.76 ± 7.33	96.10 ± 0.89	68.69 ± 5.32	60.79 ± 28.12	7.79 ± 11.02	86.16 ± 4.99	**99.82 ± 0.13**
Class 12	82.07 ± 2.43	64.14 ± 20.44	75.20 ± 3.34	**98.16 ± 1.05**	73.74 ± 14.35	30.32 ± 21.44	65.11 ± 4.43	87.50 ± 1.11
Class 13	94.27 ± 2.98	98.05 ± 1.08	96.34 ± 1.05	93.09 ± 6.11	98.50 ± 0.57	58.45 ± 23.70	94.94 ± 1.45	**98.67 ± 0.50**
Class 14	94.47 ± 0.98	92.85 ± 3.01	95.27 ± 2.17	98.90 ± 1.26	95.33 ± 4.09	71.41 ± 36.56	94.49 ± 0.62	**99.00 ± 0.29**
Class 15	95.90 ± 2.28	99.17 ± 1.15	99.53 ± 0.40	87.53 ± 2.44	99.52 ± 0.31	84.70 ± 11.50	99.11 ± 0.24	**100.00 ± 0.00**

dynamic multimodal contrastive learning module; 7) w/o Opt, removing the adaptive feature extraction and multiobjective optimization; and 8) the complete DMGNN. The final results of the ablation experiment are shown in Table IV.

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TABLE IV
ABLATION STUDY RESULTS OF DMGNN ON THE HOUSTON2013 DATASET

Metric	Baseline HSI	Baseline HSI + LiDAR	w/o LiDAR	w/o Diffusion	w/o Pre-train	w/o Opt	w/o GCL	Complete DMGNN
OA(%)	83.10 ± 0.50	95.62 ± 0.01	98.17 ± 0.07	98.59 ± 0.10	99.15 ± 0.07	99.08 ± 0.04	99.58 ± 0.04	**99.88 ± 0.02**
AA(%)	84.27 ± 0.58	94.15 ± 0.01	98.22 ± 0.16	98.90 ± 0.08	99.23 ± 0.07	99.17 ± 0.01	99.66 ± 0.03	**99.91 ± 0.02**
Kappa×100	81.70 ± 0.62	93.12 ± 0.01	98.01 ± 0.07	98.47 ± 0.11	99.03 ± 0.08	99.00 ± 0.04	99.54 ± 0.03	**99.87 ± 0.02**

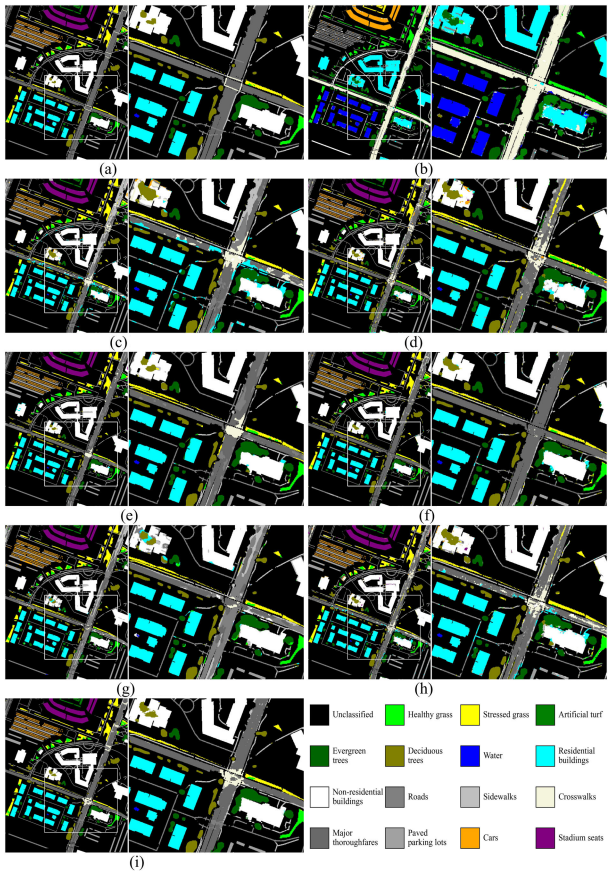


Fig. 4. Comparative experimental results on the Houston2018_sliced dataset. (a) Ground truth. (b) Bi-LSTM. (c) DSFormer. (d) Fusion_HCT. (e) GAMF. (f) MCFNet. (h) miniGCN. (i) DMGNN (ours).

1) Limitations of Single-Modality HSI and Basic GNN: The baseline (HSI) model, based solely on spectral data, produces an OA of 83.10% ± 0.50%, an AA of 84.27% ± 0.58%, and a κ of 81.70% ± 0.62%. This relatively lower performance exposes the inherent limitation of single-modality GNNs, *i.e.*, the "same spectrum, different objects" phenomenon leads to confusion in complex scenes, and the graph topology built on limited samples fails to propagate information effectively. The reason lies in that spectral similarity alone is insufficient to distinguish categories with similar material responses, while sparse labeled nodes make the neighborhood aggregation of GNNs less reliable. Interestingly, when we apply our advanced architecture without LiDAR (w/o LiDAR), the performance jumps to 98.17% ± 0.07% OA. While this significantly outperforms the basic baseline, it still lags behind the complete multimodal framework, confirming that while advanced architecture can mitigate spectral limitations, it cannot fully replace the geometric discriminability provided by auxiliary modalities. LiDAR provides complementary elevation and structural information, which assists to separate classes that are spectrally close but spatially or geometrically different.

2) Contribution of Specific Modules (Why Diffusion Outperforms Opt/GCL): As shown in Table IV, the inclusion of LiDAR data (comparing "w/o LiDAR" to "complete") yields a significant performance boost (1.71% in OA), confirming the fundamental advantage of multimodal fusion. Among the algorithmic components, the "diffusion-enhanced graph construction" emerges as the most influential module. Removing the diffusion module ("w/o Diffusion") decreases the OA by 1.29% (from 99.88% to 98.59%). In contrast, removing either the contrastive learning ("w/o GCL") or the optimization module ("w/o Opt") results in a smaller performance reduction. This disparity can be attributed to the distinct roles of the modules within the network architecture. Under limited supervision, the constructed graph typically exhibits severe topological fragmentation. The diffusion module operates at the structural level. By generating synthetic nodes, it restores the message-passing pathways. Since GNN classification depends on aggregating information from neighboring nodes, improving graph connectivity directly enhances class-consistent information propagation. Conversely, GCL and Opt function as feature-level regularizers. Although GCL aligns multimodal semantics and Opt improves boundary precision, these mechanisms rely on effective gradient propagation, which is constrained when the underlying graph topology is broken. Therefore, they mainly refine the learned representations, whereas diffusion first alleviates the structural bottleneck of the graph. Consequently, the diffusion-based augmentation provides the necessary structural foundation for the optimization and contrastive modules to operate effectively.

E. Parameter Analysis

1) Sensitivity Analysis of Hyperparameter β: The hyperparameter β plays a pivotal role in graph construction by regulating the tradeoff between spectral similarity and spatial proximity. A lower β prioritizes material consistency, while a higher β enforces spatial continuity. We evaluated the model's performance across a wide range of $\beta \in [0.1, 0.2, 0.4, 0.6, 0.8, 0.9]$ to assess the stability of the proposed DMGNN method.

As illustrated in Fig. 5, the experimental results on Houston2018_sliced dataset demonstrate that DMGNN exhibits remarkable robustness to variations in β. Despite the significant shift in graph construction logic: ranging from

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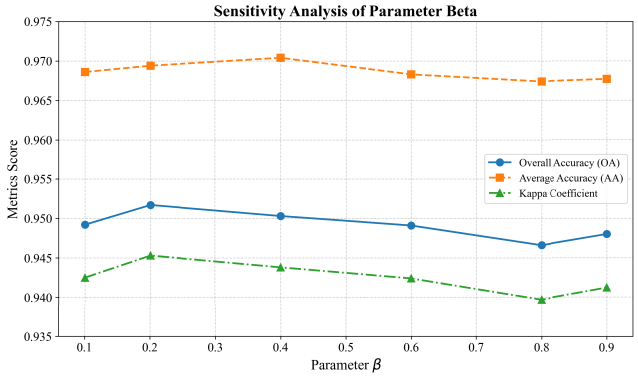


Fig. 5. Parameter analysis of hyperparameter β on Houston2018_sliced dataset.

spectrally dominant ($\beta = 0.1$) to spatially dominant ($\beta = 0.9$), the OA fluctuates within a narrow margin of less than 0.6% (from 94.66% to 95.17%). The performance peaks within the range of $\beta \in [0.2, 0.4]$, where the OA reaches 95.17% and AA reaches 97.04%, which indicates that a balanced integration of spectral features and local spatial context performs the best graph topology. Resilience to extreme settings: even at high spatial weights ($\beta \geq 0.8$), DMGNN maintains an OA above 94.6%. Similarly, at low spatial weights ($\beta = 0.1$), the model avoids severe fragmentation.

This stability can be attributed to the diffusion-enhanced multimodal graph construction and dynamic contrastive learning modules. By generating high-fidelity synthetic nodes and enforcing cross-modal alignment, DMGNN learns a feature representation that is intrinsically discriminative. Consequently, the model becomes less dependent on the precise initialization of edge weights, proving that DMGNN is not only accurate but also highly stable and easy to tune for diverse practical scenarios.

2) Sensitivity Analysis of Spectral Clustering Clusters (K_{Spec}): In the pretraining stage, the number of spectral clusters K_{spec} controls the granularity of the structural pseudo-labels. We formulate $K_{spec} = N \times C$, where C is the true number of semantic classes and N is a multiplier. To systematically determine this hyperparameter and evaluate its robustness under extreme data scarcity, we conducted a sensitivity analysis by varying N from 1 to 5 under few-shot settings (30 samples/class for Houston2013, 20 samples/class for Trento, and 70 samples/class for Houston2018_sliced), with the results shown in Fig. 6. As can be found from Fig. 6, for the Houston2013 dataset, the proposed method obtains the best OA, AA, and kappa when $N = 1$, whereas for the Trento and Houston2018_sliced datasets, $N = 3$ achieves the best performance. Therefore, $N = 1$ is optimal for Houston2013, and $N = 3$ is for Trento and Houston2018_sliced.

3) Sensitivity Analysis of Loss Function Weights (λ_{GCL} and λ_{Dice}): To explicitly justify the parameter settings of our multiobjective loss function ($\mathcal{L}_{total}$), we analyzed the sensitivity of the GCL weight (λ_{GCL}) and the dice loss weight (λ_{Dice}), as illustrated in Fig. 7. The GCL weight controls the intensity of the self-supervised structural regularization. As shown in the left panel, the OA gradually increases and peaks at $\lambda_{GCL} = 0.1$.

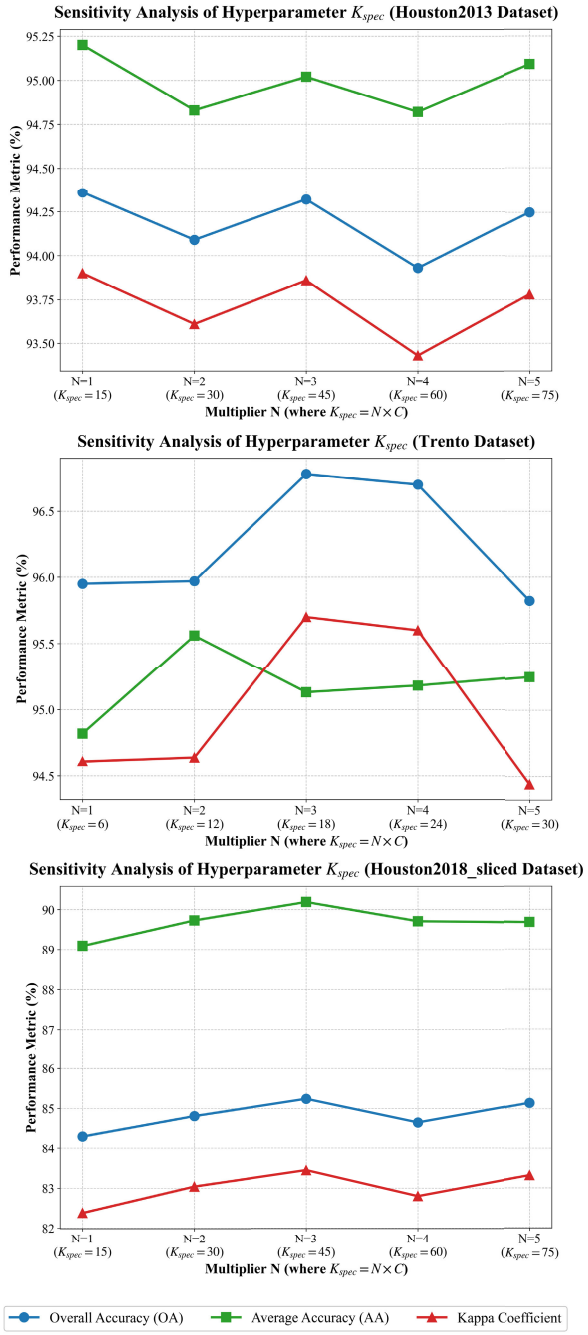


Fig. 6. Sensitivity analysis of hyperparameter K_{spec}.

When the weight is extremely low, the model fails to extract robust view-invariant features. However, when λ_{GCL} becomes excessively large (e.g., 0.5), the self-supervised contrastive penalty overwhelms the primary supervised classification gradients ($\mathcal{L}_{Focal}$), causing a slight performance degradation. Thus, $\lambda_{GCL} = 0.1$ provides the optimal regularization magnitude. Similarly, the right panel demonstrates the extreme stability of the dice loss weight (λ_{Dice}). Because the focal loss already governs pixelwise semantic correctness, the dice loss strictly focuses on macroscopic spatial overlap (IoU). The performance peaks smoothly at $\lambda_{Dice} = 0.3$, ensuring that structural region coherence is maximized without distorting the fine-grained pixel-level decision boundaries.

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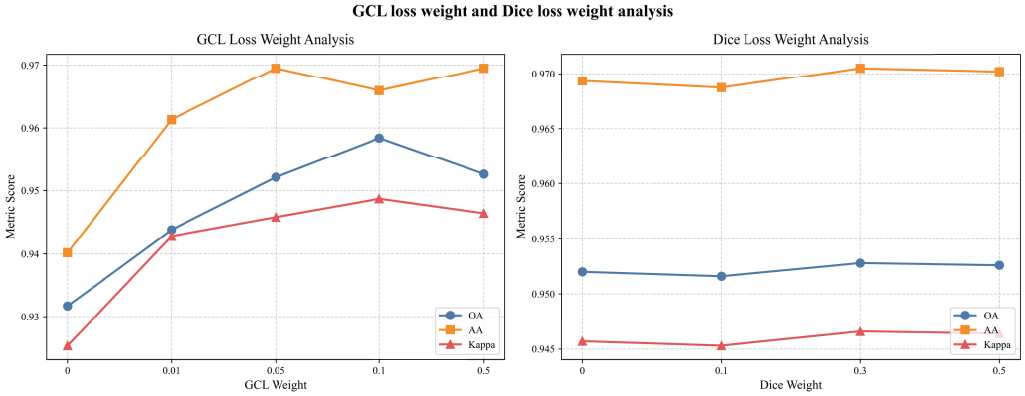


Fig. 7. Sensitivity analysis of loss function weights on Houston2018_sliced dataset.

V. Conclusion

In this work, we propose a novel DMGNN to exploit the potential of graph-structured representations in multimodal remote sensing. By integrating diffusion-based topological repair, spectral clustering-guided initialization, and dynamic contrastive alignment, DMGNN addresses the fragmented graph topology problem commonly observed under limited labeled samples. The experimental results on the Houston2013, Trento, and Houston2018_sliced datasets demonstrate that DMGNN outperforms current state-of-the-art multimodal fusion methods. These results indicate that constructing a densified graph topology from fused HSI and LiDAR data helps resolve spectral ambiguities and enables robust message passing under limited supervision. Furthermore, the ablation studies confirm the contribution of each module. The diffusion-enhanced graph construction mitigates data scarcity, while dynamic contrastive learning and multiobjective optimization improve feature robustness and boundary precision. Despite these achievements, the performance on highly complex urban scenes like Houston2018 indicates room for further refinement. Future work will focus on optimizing the computational efficiency of graph construction and exploring lightweight attention to enhance scalability for large-scale remote sensing applications.

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